

NOTATION AND TERMINOLOGY

The terminology and notation used in this book is essentially standard and borrowed from [55], if not stated otherwise; some specific notations that we shall employ are as follows:

- The category of sets and functions between them will be denoted by **Set**.
- The classifying topos of a geometric theory $\mathbb{T}$ will be denoted by **Set** $[\mathbb{T}]$ or $\mathcal{E}_{\mathbb{T}}$.
- $\gamma_{\mathcal{E}} : \mathcal{E} \rightarrow \mathbf{Set}$ will denote the unique (up to isomorphism) geometric morphism from a Grothendieck topos $\mathcal{E}$ to the category **Set** of sets.
- By a cartesian category (resp. cartesian functor) we mean a category with finite limits (resp. a finite-limit-preserving functor).
- Given a category $\mathcal{C}$, we shall often write $c \in \mathcal{C}$ to mean that c is an object of $\mathcal{C}$.
- The collection of objects (resp. arrows) of a category $\mathcal{C}$ will be denoted by $\mathrm{Ob}(\mathcal{C})$ (resp. $\mathrm{Arr}(\mathcal{C})$).
- The category of functors from a category $\mathcal{C}$ to a category $\mathcal{D}$ will be denoted by $[\mathcal{C}, \mathcal{D}]$.
- The terminal object of a category $\mathcal{C}$, if it exists, will be denoted by $1_{\mathcal{C}}$, or simply by 1 .
- The initial object of a category $\mathcal{C}$, if it exists, will be denoted by $0_{\mathcal{C}}$, or simply by 0 .
- The identity arrow on an object c in a category will be denoted by 1_c .
- By a subterminal object in a category $\mathcal{C}$ having a terminal object 1 we mean an object c such that the unique arrow $c \rightarrow 1$ in $\mathcal{C}$ is a monomorphism.
- In a category $\mathcal{C}$ with a terminal object 1 , for any object c of $\mathcal{C}$, we denote by $!_c : c \rightarrow 1$ the unique arrow in $\mathcal{C}$ from c to 1 .
- In a category with finite products, we shall denote by $\langle f_1, \dots, f_n \rangle$ the arrow $c \rightarrow a_1 \times \dots \times a_n$ determined by the universal property of the product by a collection of arrows $f_i : c \rightarrow a_i$ (for $i = 1, \dots, n$).
- Given an object c of a category $\mathcal{C}$ and a category $\mathcal{I}$, we shall denote by Δ_c the constant functor $\mathcal{I} \rightarrow \mathcal{C}$ with value c .
- We shall say that a category is small if it has only a set of objects and arrows, and that a category is essentially small if it is equivalent to a small category.
- The (meta-)2-category of (large) categories, functors and natural transformations between them will be denoted by $\mathfrak{CAT}$.

- The 2-category of small categories, functors and natural transformations between them will be denoted by $\mathbf{Cat}$.
- The 2-category of Grothendieck toposes, geometric morphisms and geometric transformations between them will be denoted by $\mathbf{BTop}$ and, given two Grothendieck toposes $\mathcal{E}$ and $\mathcal{F}$, $\mathbf{Geom}(\mathcal{E}, \mathcal{F})$ will denote the category of geometric morphisms $\mathcal{E} \rightarrow \mathcal{F}$ and geometric transformations between them.
- Given a Grothendieck topos $\mathcal{E}$ and a category $\mathcal{C}$, we will write $\mathbf{Flat}(\mathcal{C}, \mathcal{E})$ for the category of flat functors $\mathcal{C} \rightarrow \mathcal{E}$ and natural transformations between them. Given a Grothendieck topology J on $\mathcal{C}$, $\mathbf{Flat}_J(\mathcal{C}, \mathcal{E})$ will denote the full subcategory of J -continuous flat functors $\mathcal{C} \rightarrow \mathcal{E}$.
- Given a geometric theory $\mathbb{T}$, we shall denote by $\mathcal{C}_{\mathbb{T}}$ the geometric syntactic category of $\mathbb{T}$ and by $J_{\mathbb{T}}$ the canonical (Grothendieck) topology on $\mathcal{C}_{\mathbb{T}}$.
- Given a theory of presheaf type $\mathbb{T}$, we shall denote by $\mathbf{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ the category of (representatives of isomorphism classes of) finitely presentable models of $\mathbb{T}$ in $\mathbf{Set}$.
- By a poset we mean a partially ordered set.
- Given a locally small category $\mathcal{C}$ and an object c of $\mathcal{C}$, the covariant (resp. contravariant) representable functor associated with c will be denoted by $\mathcal{C}(c, -)$ or $\mathrm{Hom}_{\mathcal{C}}(c, -)$ (resp. by $\mathcal{C}(-, c)$ or $\mathrm{Hom}_{\mathcal{C}}(-, c)$).
- A sieve generated by a single arrow f in a category will be denoted by (f) .